

A Full Metro-scale Time-Dependent Traffic Route Model

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ABSTRACT

We describe how to train an urban traffic model on a full metropolitan road network, up to 5-minute resolution on a week-long model. The training data is from GPS probes representing about 5% of all traffic in that area over a 30-day period. Building such a large model (about 100,000 links, and 2,016 time windows for a week), presents many challenges. First, the 5% of all traffic (about 2.5 million trips) is still incredibly sparse on this scale. This means it is essential to smooth out the data and error evaluation, and also not learn too many parameters at risk of over-fitting. Second, because the scale is so large, to be able to both train parameters and also generate feasible routes and not just static distributions, it is important we learn a PageRank-style model. We demonstrate the model's ability for route forecasting around three changes in traffic behavior.

CCS CONCEPTS

• **Computing methodologies** → **Modeling and simulation**; Spatial and physical reasoning; • **Information systems** → *Geographic information systems*.

KEYWORDS

PageRank, road traffic, GPS trajectory, map matching, hidden Markov model, graph diffusion, urban mobility, per-link traffic estimation

ACM Reference Format:

Hamid Shafieasl, Xiaoyue Cathy Liu, and Jeff M. Phillips. 2026. A Full Metro-scale Time-Dependent Traffic Route Model. In *Proceedings of 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*. ACM, New York, NY, USA, 13 pages. <https://doi.org/XXXXXX.XXXXXX>

1 INTRODUCTION

In this paper we design a full metro-scale traffic model with the following objectives: (1) It should consider traffic on each link of a road network. Otherwise, it is not compatible with low-level route planning, actual map-matched GPS-traces. (2) It should be temporal, with a small resolution like 5 minutes, and spanning a week. Traffic patterns are variable at this resolution level, due to commute and meal regularity. The week long models are necessary for regular changes in behavior over that period. (3) It should be able to *efficiently* produce actual road-network routes mirroring GPS traces as well as global traffic population distributions directly,

starting in each time window. These are necessary for producing realistic routes that can adapt to the environment for planning and forecasting. The efficiency is necessary so this can be deployed in other AI infrastructure for generative modeling. (4) Finally, it should learn from and be trained on actual data so it can represent the true human behaviors and adapt if new data arises.

We build such a system on the greater Salt Lake City metropolitan area. This represents a cohesive and connected ecosystem with about 100,000 links. We train this model on GPS tracing representing about 5% of the traffic for the area for a 30 day period (about 2.5 million routes). The scale of the data is large, but manageable by today's standards – the real challenges is the sparsity. After filtering and cleaning, only about 180 million GPS points remain; but at a 5-minute resolution with 2,016 time periods over a week, there are over 800 million links×time-windows to model. Even if we had a longer observation period or a higher percentage of routes, a similar challenge would remain since many routes are concentrated on busy thoroughfares and are peak times of the day. The tail of density is extremely long.

We manage to effectively build a model satisfying this desiderata through several innovations. Our baseline is a PageRank model [?] that treats directed network links as graph vertices, and intersections model outgoing edges. We then categorize links by speed and number of lanes, and use data to learn transition probabilities between edges that favor transitions to or from higher throughput roads. However, routes typically move towards thoroughfares as they depart, and then move away from them as they reach their destination; so we design a two-phase PageRank that has separate transition probabilities as it gears up, and then speeds down. Finally, when a route ends, PageRank “teleports” to a random node to start the next walk; however, this does not reflect the pattern of route origins. So we use our observations of route starting points to also inform this teleportation probability, akin to trust-rank or personalized models [?]. We demonstrate that each of these steps is important towards building an effective learned model.

Another challenge is in a trustworthy evaluation. Due to the data sparsity, even without a cross-validation split, it is not enough to directly predict the link density accuracy. Across the 4 full weeks, only 3% of links×time have at least one observation in a week, and only 1% have at least one observation in more than one week. So we disperse the sparse observations we have using graph diffusion [9]. We show how to apply this in a controlled way so that for each full week of observation, each link×time has non-zero measure, and the average relative change in value is less than the median number of difference in observations among the links×time which have observations in more than two weeks. That is, this diffusion makes the observations dense in a natural way that does not disrupt the data below the noise floor. From here, we can evaluate the model by training on one week, and measuring on the next.

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SIGSPATIAL '26, November 3–6, 2026, Riverside, CA, USA

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ACM ISBN 978-1-4503-XXXX-X/2026/11
<https://doi.org/XXXXXX.XXXXXX>

Ultimately, we demonstrate an efficient, fine-grain, and full model at metro scale. It produces natural temporally encoded routes, and its error is on par with the week-to-week variability of the diffused observations. Importantly, because it is a PageRank model, we can use power iterations to compute the limiting global distribution for any time window, and are not reliant on noisy 1-by-1 forward simulations. Moreover, we demonstrate on three scenarios (a large sporting event, a street festival, and an accident-induced lane closure) how we can use it to effectively forecast heavy traffic and investigate what-if scenarios. PageRank’s limiting traffic distribution can quickly adapt to these scenarios.

2 RELATED WORK

This paper designs a method to fit a calibrated model of vehicle traffic at the resolution of *every single road link* of an entire city ($\sim 100,000$ links) at 5-minute temporal resolution over a week, using probe trajectories on a fixed road network. The closest prior work falls into three groups: agent-based traffic simulators that take demand as an input rather than fit it from data [2, 3]; origin-destination estimators that operate at a coarser zone or node resolution [4, 11]; and PageRank-style random walks applied to urban networks for centrality or coarse congestion prediction [1, 5, 10].

The closest comparable effort in scope is MATSim [2, 3], a long-running agent-based simulator that, given a synthetic population, schedules, and a network, performs *forward* simulation of per-link traffic at every time of day – or similarly for more local models like SUMO [?] or DTA [?]. Our problem is the *inverse*: given observed probe trajectories, infer a per-link popularity field and validate it against held-out weeks. The two efforts complement each other; we use the MATSim network model of Salt Lake City as our graph, but we can directly determine the limiting traffic distribution.

A long line of transportation work estimates city-wide origin-destination (OD) matrices from cellular records combined with link counts [4] or from sparse probe trajectories combined with link counts [11]. These methods produce an OD matrix at zone or node resolution: each entry quantifies how many trips begin in zone *a* and end in zone *b*, but commits to no particular route in between. Our learned per-link traffic field is finer modeling unit than OD estimation, and also incorporates temporal modeling.

PageRank has been adapted to urban networks in a variety of ways complementary to ours. It has been used for structural centrality ranking of links in the network [1], or other ranking of urban spaces [5]. It has been used to model congestion, for instance by Wang *et al.* [10] on a coarse 4,000 cell grid of Beijing. These methods predominantly use PageRank for analysis, and may even evaluate it against some observations, but it is not trained on full traffic observations, and not at the level of link or time granularity that we build. Our higher-resolution model, fit to real observations, will provide a better setting on which to perform these sorts of analysis.

3 DATA AND PREPROCESSING

Our raw input is a corpus of sparse GPS trajectory records collected by INRIX, a commercial provider of probe-based vehicle movement and traffic data. It covers the Salt Lake City metropolitan area for September 2018, a continuous 30-day observation window. The data are distributed as a set of data rows, with ones belonging to the

same device are interleaved chronologically. Each row corresponds to a single GPS fix and contains four fields:

Column	Meaning
did	anonymized device identifier (string)
ts	timestamp, formatted YYYY-MM-DD HH:MM:SS
lat	latitude of the GPS fix (decimal degrees)
lon	longitude of the GPS fix (decimal degrees)

The raw corpus contains 924,379,163 GPS locations; however roughly 80% of them fall outside the bounding box around the Salt Lake City metro area we study (see Figure 1), and are dropped at the first preprocessing stage, leaving 179 million locations inside the metro area. After full preprocessing and the map matching of Section 3.1, the corpus is reduced to approximately 2,477,000 distinct trips, of which about 714,594 survive map matching and contribute to the population matrix. The remaining trips are discarded for one of three reasons: fewer than five GPS locations, locations that fall outside the road network, or no viable Viterbi path (from map matching). At this stage the data are purely geographic point clouds: the observations are not yet associated with any road link, and no network structure has been introduced.

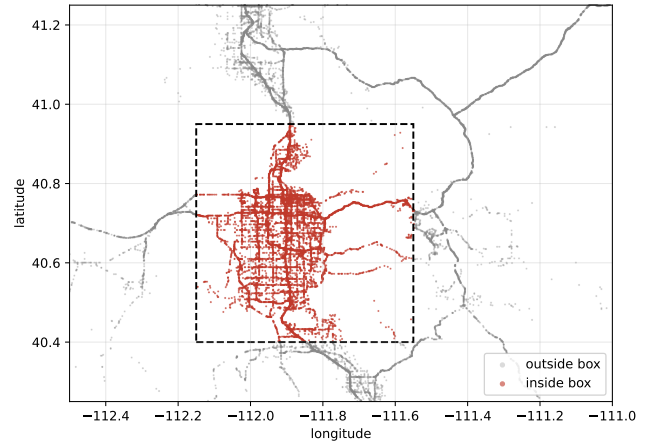


Figure 1: Raw GPS locations (random sample of $\sim 10^5$ from 924 M) and the SLC bounding box. Red points fall inside the box (40.4–40.95°N, 112.15–111.55°W, dashed); gray fall outside.

3.1 Map-Matching

We project the raw GPS fixes onto the road network with the Leuven implementation [6, 7] of the hidden Markov map-matcher of Newson and Krumm [8]. The matcher models the GPS-to-link perpendicular distance as a zero-mean Gaussian with standard deviation σ (in meters); σ is the only parameter and is corpus-dependent. Newson and Krumm fit $\sigma \approx 4.07$ m on their Seattle corpus; on our sparser SLC corpus we refit to be self-consistent. We re-run the matcher across a range of candidate σ (with a candidate pruning radius $\text{max_dist} = 400$ m, the matcher’s default upper bound on emission distance), compute the standard deviation of the per-fix residuals over the surviving fixes, and pick the smallest σ at which the declared emission std is no longer smaller than the measured one.

Table 1: Self-consistency check for the GPS noise scale σ on the SLC corpus.

σ_{declared}	σ_{std} (filtered)	gap = $\sigma_{\text{std}} - \sigma$	verdict
1.00	2.20	+1.20	too tight
1.75	2.77	+1.02	too tight
3.00	3.15	+0.15	slightly too tight
3.50	3.22	-0.28	first $\sigma > \sigma_{\text{std}}$ (chosen)
3.75	3.25	-0.50	too loose
4.00	3.26	-0.74	too loose
5.00	3.31	-1.69	too loose

The sign of $\sigma_{\text{std}} - \sigma_{\text{declared}}$ flips between $\sigma = 3.0$ and $\sigma = 3.5$. We therefore set $\sigma = 3.5$ m throughout this paper.

Two map versions. The pipeline operates on two views of the same SLC road network. The *low-resolution* map is the MATSim simulator network ($N \approx 99,716$ directed links). The *high-resolution* map is the same OSM extract with long or curved MATSim links subdivided into shorter sub-edges so that arterial geometry is captured by collinear sub-edges. The main pipeline uses high-resolution matching aggregated to low-resolution links; the low-resolution baseline uses low-resolution matching instead.

3.2 Population Matrix

The final stage of the pipeline aggregates the matched edge visits into a sparse matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ that is the input to the rest of the model. Each row corresponds to a time bin, each column to a road link, and each entry counts the number of edge visits that fell into that bin and link.

Time is partitioned into non-overlapping bins of equal width $\Delta = 300$ seconds (five minutes). For an absolute timestamp t measured in seconds since the epoch $t_0 = 2018-01-01$, the bin index is $b(t) = \lfloor \frac{t-t_0}{\Delta} \rfloor$. There are $T = 30 \times 24 \times 12 = 8,640$ five-minute bins. A five-minute resolution is chosen to balance two competing constraints: too coarse a bin (e.g., one hour) blurs the rush-hour profile of the network; too fine a bin (e.g., one minute) leaves the matrix excessively sparse, with most cells empty even on busy links.

Deduplication. The map-matched output of Section 3.1 provides, for each trip r , a list of linkID (lid) - time pairs (lid_j, t_j) , one per emitting GPS location. The counting rule is:

- (1) **Bin assignment.** Each pair is assigned to the bin

$$b_j = b(t_j) = \left\lfloor \frac{t_j - t_0}{\Delta} \right\rfloor.$$

- (2) **Per-trip deduplication.** For a given trip r , all fixes that fall in the same bin and on the same link are collapsed into a single contribution. Concretely, the set of contributions from trip r is the set of distinct triples

$$\{ (b_j, \text{lid}_j, r) : j = 1, \dots, M_r \}.$$

If the same trip emits five GPS points that all map to the same bin and the same link, they are counted exactly once.

- (3) **Accumulation.** The population-matrix entry $C_{b,i}$ is then the number of distinct trips that contributed at least one location

to bin b on link i :

$$C_{b,i} = |\{ r : \exists j \text{ with } b_j = b, \text{lid}_j = i \}|. \quad (1)$$

The per-trip deduplication ensures that $C_{b,i}$ measures link occupancy by distinct vehicles in the corpus, not the GPS-emission density of those vehicles: a trip that produces several GPS points while crossing one link inside a single 5-minute window contributes one unit, not several.

Sparse matrix representation. The matrix C has dimensions $T \times N \approx 8.6 \times 10^8$ cells, where $T = 8,640$ is the number of five-minute bins (Section 3.2) and $N = 99,716$ is the number of directed road links. Only a small fraction of these cells are nonzero: at five-minute resolution most road links carry no trip in any specific window. We therefore store C in compressed sparse row format alongside the per-row timestamps and per-column link identifiers, which on the SLC corpus reduces the artifact to roughly 7 MiB.

Limitations of C from Bias and Sparsity. Two limitations are worth stating up front. First, the corpus is a biased sample of real traffic: it is drawn from a specific population of GPS-emitting devices and is not a uniform sample of all vehicles on the road. In particular, it biases towards commercial vehicles, with 42% of raw waypoints belonging to trucks. Links that systematically attract devices in the sample (e.g. commercial freeway corridors) are over-represented relative to links that attract a different population (e.g. residential cul-de-sacs). The matrix should therefore be read as a *prior* on link usage, not as ground-truth volume. We note that while our INRIX sample has its biases, this is likely to be also true, but with different bias, for other entities capably of collecting data: e.g., Uber drivers, Google Maps users, Amazon deliveries, Tesla cars.

Second, even within the large corpus, sparsity is structural: at any single five-minute slot, the median number of GPS fixes per link is zero, and only a few hundred to a few thousand links carry any matched visit at all. To quantify this at the bin-of-week level, we aggregate the first four full weeks of the corpus (8,064 bins, i.e. $4 \times 7 \times 288$ five-minute slots) and ask, for each (*link, bin-of-week*) cell, in how many of the four weeks it received at least one matched fix. The result is in Table 2.

Table 2: How often each (link, bin-of-week) cell was zero across the four weeks. Active means $k \neq 4$.

weeks with zero count	cell count	fraction
$k = 0$ (active in every week)	304,983	0.15%
$k = 1$	499,625	0.25%
$k = 2$	1,153,583	0.57%
$k = 3$	3,994,889	1.99%
$k = 4$ (never observed in 4 weeks)	195,074,376	97.04%
total	201,027,456	100%

We say a link for a bin-of-week is *active* if for at least one week, at the specified time it has non-zero traffic. 97% of all (link, bin-of-week) cells are zero in every one of the four weeks; only 0.15% are active in all four. Most active cells are active in only one of the four weeks, i.e., represent a single noisy sample rather than a stable rate.

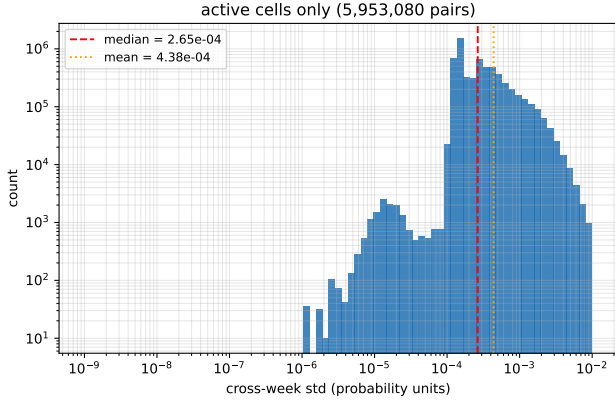


Figure 2: Distribution of the cross-week standard deviation $\tau_{b,i} = \text{std}_w E_b^{(w)}(i)$ across the four weeks, per (link, bin-of-week), restricted to the 5.95×10^6 active cells.

4 BUILDING THE TRAFFIC POPULATION MEASURE

The matrix C is converted into a per-bin traffic population prior by row-normalization. For every bin b with $\sum_i C_{b,i} > 0$, define

$$F_b(i) = \frac{C_{b,i}}{\sum_{i'} C_{b,i'}}.$$

This makes F_b a probability distribution over the N links of the network: the share of trip-presence mass during bin b that landed on link i . Time bins with no observed trips are assigned the uniform distribution $F_b(i) = \frac{1}{N}$, which acts as an uninformative fallback.

Sparsity and Noise Floor. The traffic population prior F_b summarizes what was actually observed on the road network during time bin b , but it has two structural shortcomings for predictive use. First, F_b is sparse: a typical 5-minute window puts non-zero mass on only a few thousand of the N links, leaving the rest of the network at exact zero. Second, the empirical signal is noisy at fine resolution: the same heavy corridor can be touched by zero observed trips in one time bin and several trips in the very next, simply because no GPS device happened to emit a location on that link inside that 5-minute slot.

We measure the scale of this noise from the four-week structure of the corpus. For each (link, bin-of-week) cell that the corpus saw at least once, we compute the standard deviation $\tau_{b,i}$ of its normalized count $F_b(i)$ across the four weeks. This captures how much the same link in the same time-of-week jitters from week to week.

The median of $\tau_{b,i}$ over the 5.95×10^6 active cells in Table 2 is $\tau_{\text{global}} = 2.65 \times 10^{-4}$, a single corpus-wide scalar *noise floor*, used as a global yardstick of how much variation to trust. See Figure 2, which shows a log-log plot of count of these $\tau_{b,i}$ values. Differences in F_b much smaller than τ_{global} are not clearly distinguishable from week-to-week sampling noise. We use this noise floor to calibrate the smoothing step below.

4.1 Diffusion Smoothing of Population Prior

A graph-aware smoothing step addresses both issues at once. It lifts every link to a positive value (so the prior never has hard zeros) and borrows mass from neighboring links (so a heavy link does not disappear because of an empty bin). The smoothing is applied row by row to the count matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ in two steps.

Step 1. Laplace pseudo-count. For every time bin b and every link i , add a small factor $\alpha = \frac{1}{100}$ to the raw count

$$c_b(i) = C_{b,i} + \alpha, \quad \alpha = \frac{1}{100}. \quad (2)$$

After this step every entry of c_b is positive: the row is dense before any further processing.

Step 2. Single-step graph diffusion. Let $P \in \mathbb{R}^{N \times N}$ be the row-stochastic adjacency matrix of the directed road graph: $P_{i,j} = \frac{1}{d_i^{\text{out}}}$ when there is a road link from i to j , and zero otherwise; here d_i^{out} is the out-degree of link i . Apply one random-walk step on the graph,

$$\tilde{c}_b = (1 - \gamma) c_b + \gamma P c_b, \quad \gamma = 0.20, \quad (3)$$

where $(P c_b)_i = \sum_j P_{i,j} c_b(j)$ is the average of $c_b(j)$ over the out-neighbors j of link i . The mass on link i is therefore averaged with the mass on the links it leads to. Finally, normalize each row to a probability distribution:

$$\tilde{F}_b(i) = \frac{\tilde{c}_b(i)}{\sum_j \tilde{c}_b(j)}. \quad (4)$$

The result $\tilde{F}_b \in \mathbb{R}^N$ is the *diffused traffic population*: dense, positive on every road link, and preserving the order of the heaviest links across the corpus. $\tilde{F}_b$ is used as an alternative to the raw F_b throughout the evaluation of ??.

4.2 Choosing γ from the Noise Floor

We pick the diffusion strength γ so that the smoothing rarely displaces any single cell by more than about half of that cell's own week-to-week sampling noise. Concretely, for each active (link i , bin-of-week b) cell — that is, with at least one observed positive count across the four full corpus weeks $w = 1, \dots, W$ (with $W = 4$) — define the per-week displacement

$$D_{b,i}^{(w)}(\gamma) = \left| \tilde{F}_{b,i}^{(w)}(\gamma) - F_{b,i}^{(w)} \right|,$$

as the difference between the smoothed normalized count $\tilde{F}_{b,i}^{(w)}(\gamma)$ with parameter γ , and the observed count in that week $F_{b,i}^{(w)}$. The cell-level displacement averaged across the four weeks

$$D_{b,i}(\gamma) = \frac{1}{W} \sum_{w=1}^W D_{b,i}^{(w)}(\gamma),$$

and compare it to the cross-week noise floor $\tau_{b,i}$ of the same cell or the global noise floor τ_{global} . In particular, we aim for

$$\text{per-cell: } D_{b,i}(\gamma) \leq \frac{1}{2} \tau_{b,i},$$

$$\text{global: } D_{b,i}(\gamma) \leq \frac{1}{2} \tau_{\text{global}} = 1.325 \times 10^{-4}.$$

Table 3 shows statistics on this for various choices of γ . It plots the median ratio $D_{b,i}(\gamma)/\tau_{b,i}$ or $D_{b,i}(\gamma)/\tau_{\text{global}}$ (as med.). It also provides the 95th percentile of that ratio, and the percent of cells

Table 3: Quantiles of the dimensionless displacement-to-noise ratio $D_{b,i}(\gamma)/\tau$ across the 5,954,929 active SLC cells, and percentage that exceed 0.5 ratio criteria. Left block uses the per-cell denominator $\tau_{b,i}$, right block uses the global denominator $\tau_{\text{global}} = 2.65 \times 10^{-4}$.

γ	per-cell criterion			global criterion		
	med.	95th pct.	> 0.5	med.	95th pct.	> 0.5
0.00	0.000	0.000	0.00%	0.000	0.000	0.00%
0.05	0.038	0.100	0.42%	0.034	0.215	0.90%
0.10	0.069	0.186	1.00%	0.062	0.393	3.31%
0.12	0.081	0.219	1.27%	0.073	0.463	4.40%
0.13	0.087	0.236	1.41%	0.079	0.498	4.96%
0.14	0.093	0.253	1.56%	0.084	0.533	5.53%
0.15	0.099	0.270	1.73%	0.090	0.568	6.10%
0.20	0.130	0.354	2.62%	0.117	0.741	8.95%
0.25	0.160	0.436	3.63%	0.144	0.912	11.87%
0.30	0.190	0.518	5.39%	0.171	1.081	14.81%
0.40	0.248	0.680	9.42%	0.224	1.413	20.64%
0.50	0.306	0.838	17.29%	0.276	1.737	26.59%
0.70	0.418	1.145	32.10%	0.376	2.364	37.83%

which exceed the 0.5 criteria. The per cell criteria fairs better than the global one, indicating that error decreases with these population rates. We choose $\gamma = 0.2$ since it provides sufficient smoothing, but violates the criteria a small fraction of the time.

Violator tail at $\gamma = 0.20$. To see how far past the tolerance the violators sit, we sweep the threshold η in $D_{b,i}(0.20) > \eta \tau_{\text{global}}$ across the active cells:

η	0.2	0.5	0.8	1.0	1.5	2.0
fraction $> \eta \tau_{\text{global}}$	28.5%	8.9%	4.4%	3.0%	1.3%	0.6%

The violator tail thins quickly: of the 8.9% that exceed half the noise floor, only 3% exceed τ_{global} and fewer than 0.7% exceed $2 \tau_{\text{global}}$, so the smoothing rarely changes a cell by more than its own week-to-week jitter.

4.3 Evaluation Mechanism

Now given the diffused population prior $\tilde{F}_b$, which we use as our ground truth proxy $t_i = \tilde{F}_b(i)$, we set up how we measure our model. Since the population of each cell still varies significantly at different scales, we use a relative error for a prediction p_i at a time bin i is,

$$r_i = \frac{|t_i - p_i|}{t_i + \varepsilon},$$

where $\varepsilon = 10^{-6}$ is included to add stability. Note that this ε is less than $\tau_{\text{global}}/100$, less than the effect of the $\frac{1}{10}$ th of a trip on the busiest cell. Then the *overall score* at a time bin b is the mean relative error

$$\text{MRE} = \frac{1}{N} \sum_{i=1}^N r_i.$$

One week to next forecast. We frame our predictions by using data from one week to predict (or forecast) the next. This is similar to the noise floor, where we calibrated against the raw data set statistics predicting the next week at the same link and same

time/day. For our prediction, we use a forecast for a model built on one week, predicting to the same time period exactly 7 days forward (move forward time index by 2016). For global analysis, we train parameters on the first 23 days of the dataset, aggregate them to the days/times of a single week, and then predict on the final seven days of the week, averaged across all 2,016 of the 5 minute time windows.

5 PAGERANK MODEL FOR LINK-LEVEL TRAFFIC

We now describe our PageRank model for representing traffic patterns on our metro-scale network. Recall that PageRank [?] was developed as a model of a random surfer on the internet; the resulting distribution of where the random surfer would visit in the limit formed the basis for Google's groundbreaking search engine. In this way it models the outgoing links of a webpage as directed edges in a graph, with pages as N nodes; these form the probability transition matrix $P \in \mathbb{R}^{N \times N}$. The i th column $P_i \in \mathbb{R}^N$ is a probability distribution for node i where $P_i[j]$ encodes the probability that a random walk at node i will be in node j in the next step. In basic PageRank, these probabilities are set uniformly among the outgoing hyperlinks of a webpage, and P is very sparse.

Just using P does not define an ergodic Markov Chain, so a key innovation is having the random surfer *teleport* to a random webpage with some probability p (usually about 0.15). This makes the random walks follow a geometric distribution in length, with expected length $1/p$ (about 7 steps on average). It also makes the true transition probability dense, but still allowing efficient simulation of forward steps from initiate state v_n as:

$$v_{n+1} = (pE + (1-p)P)v_n. \quad (5)$$

Here $E \in \mathbb{R}^{N \times N}$ represents the uniform random distribution, so it is the $1/N$ times the all 1 matrix. However, note that this step can be implemented by replacing pEv_n with a fixed vector $e \in \mathbb{R}^N$ that has p/N in each coordinate.

5.1 Baseline Traffic PageRank Model

Our baseline traffic model is to adopt this PageRank model to our large roughly 100,000 ($N = 99,716$) link network. Since our population data mapped to the road network links, these links map to the *nodes* in the graph, not the edges. The intersections (and direct connection between consecutive links) are the edges. This switch (links $\neq$ edges) is slightly un-intuitive, but allows for the precise sort of modeling we seek, and aligns with our population data mapping to links.

Then we can define the probability transition matrix $P \in \mathbb{R}^{N \times N}$ by the road network; we make the probability transition uniform $1/k$ among all (k) options from a directed link. Then with $E \in \mathbb{R}^{N \times N}$ with each entry $E_{i,j} = 1/N$ our transition update follows (5) precisely.

In addition to this modeling being able to generate realistic routes, it can also generate global traffic distributions. By iteratively applying (5), the limit $\lim_{n \rightarrow \infty} v_n = v_\star$ provides the modeled distribution of traffic across all road network links.

Calibrating p from the trip corpus. We next need to calibrate p from the trip corpus. The chain in (5) is a geometric absorption

process: the number of graph steps before a teleport is $\text{Geom}(p)$ on $\{0, 1, 2, \dots\}$ with mean $\frac{1-p}{p}$. We match p to the empirical mean trip length $\mathbb{E}[K_{\text{total}}]$ by method-of-moments:

$$p = \frac{1}{1 + \mathbb{E}[K_{\text{total}}]}.$$

On the map-matched corpus (714,594 trips, all route lengths), the empirical mean is $\mathbb{E}[K_{\text{total}}] = 18.68$, which gives $p = 0.0508$. We thus use this for this $p = 0.0508$ going forward. Figure 3 shows the empirical K_{total} histogram with the $\text{Geom}(p = 0.0508)$ reference overlaid, illustrating the nice geometric fit to true route length distribution.

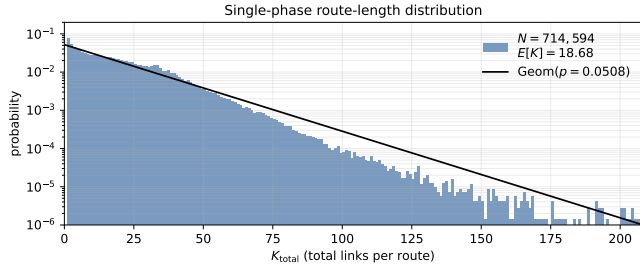


Figure 3: Empirical histogram of trip length K_{total} on the 714,594-trip corpus, with the $\text{Geom}(p = 0.0508)$ overlay.

Implementation Details. The sparse probability transition probability matrix P , which as with the map-matched observation matrix C , we represent as a sparse matrix. It describes the directed road graph: each column of P being uniform over the outgoing road neighbors of one link. Then using a teleport probability $p = 0.0508$, the iteration in (5) is applied until $\|v_{n+1} - v_n\|_1 < 10^{-7}$, with a safety cap of $n = 300$ iterations. The cap is conservative: the power iteration converges geometrically at rate of at least $1 - p$, so the theoretical worst-case number of iterations to reach L1 error 10^{-7} is

$$n^* = \frac{\ln 10^{-7}}{\ln(1-p)} = \frac{\ln 10^{-7}}{\ln 0.9492} \approx 310.$$

We checked empirically on 1,500+ random target bins and the chain converged before 300 iterations in every single case; the observed maximum was 254 iterations and the median was 209.

5.2 PageRank Model with Informed Teleportation

Beyond this baseline model, the first modification we make is so that the teleportation step can be more informed than the default uniform. That is, which links to start routes are not uniform. Its quite rare to *start* a route on a busy freeway, and it should be more likely to start one from a grocery store parking lot.

For each time window b we can track all routes that start in that period. After normalizing these counts we can get a probability distribution $e_b \in \mathbb{R}^N$. Like the traffic population matrix F_b , this also faces an extreme sparsity challenge. So we apply the same graph diffusion method (as in Section 4.1) to get a diffuse route initialization distribution $\tilde{e}_b$.

Our first innovation replaces the uniform random teleportation matrix with the population diffusion model of routes $\tilde{E}_b$, where each column is $\tilde{e}_b$. Then the simulation update step is

$$v_{n+1} = (p\tilde{E}_b + (1-p)P)v_n = p\tilde{e}_b + (1-p)Pv_n. \quad (6)$$

This accomplishes two important aspects. First, it adapts the simulation to start routes at more realistic locations informed by actual data. Second, allows the model to adapt to the specific time-window b ; so the traffic model can generate different routes at different times during the day or week.

5.3 PageRank Model with Learned Transitions

We next apply machine learning to learn a more informed probability transition matrix P . A potentially unrealistic part of a PageRank model is that the routes are random walks, while realistic routes are directed and meaningful. However, the Markov chain modeling requires the *Markov* properties that each step is memoryless, so we cannot use this entire set up if we inform an origin-destination pair. However, we can adapt the transition probabilities to reflect more realistic to choices that model traffic. We do so with two modeling aspects following the principals that routes tend to move towards higher-throughput links.

These aspects leverage physical throughput attributes of links ℓ of the graph: flow speed $s(\ell)$ and a lane count $\lambda(\ell)$. Let $\bar{s}(\ell) = s(\ell)/\bar{s}$ and $\bar{\lambda}(\ell) = \lambda(\ell)/\bar{\lambda}$ denote these attributes normalized by their per-link mean over the network.

Now given from a link ℓ_i the weight of each potential outgoing link ℓ_j is parameterized by 2 parameters α_s and α_l so that

$$w(j) = \bar{s}(\ell_j)^{\alpha_s} \bar{\lambda}(\ell_j)^{\alpha_l}.$$

Finally, we can set the transition probability matrix from i to j among all outgoing edges J_i from i as

$$P_{j,i} = \frac{w(j)}{\sum_{j \in J_i} w(j)}.$$

And P retains its sparsity with $P_{j,i} = 0$ for $j \notin J_i$.

It remains to set the values for $\alpha = (\alpha_s, \alpha_l)$. These values α_s, α_l are larger when there is a preference to turn onto a link with higher throughput (because of higher flow-rate e.g., speed limit, or number of lanes). A negative value in α_s, α_l would indicate a preference to turn onto smaller roads.

Towards determining these, we set up a loss function

$$\mathcal{L}(\alpha) = \sum_{b \in [T]} \sum_{i \in [N]} \frac{|v_\star(\tilde{E}_b, \alpha)[i] - \tilde{F}_b[i]|}{\tilde{F}_b[i] + \epsilon}$$

where $v_\star(\tilde{E}_b, \alpha)$ is the converged-to PageRank vector for a Markov chain following (6), where P is defined using the specified α parameters. Given the diffused starting positions $\tilde{E}_b$, the goal is to match the diffused observed traffic population $\tilde{F}_b$. It is the MRE (mean relative error), summed over all time periods b ; thus the goal is to set these α parameters globally for all timescales; that is to learn the driving preferences for the metro area regardless of the time of day. While it may be the case that these driving patterns change throughout the day or week, we opt to learn a smaller set of parameters with goal of not overfitting. Moreover, this training step is somewhat computationally expensive, and repeating it 8000+ times, would be exorbitant.

Finally, we can choose these parameters $\alpha^* = \arg \min_{\alpha} \mathcal{L}(\alpha)$. We do this with gradient descent. We use autograd to determine the gradient, and try several random restarts to confirm it reliably finds the same minimal parameter choices.

5.4 Two-phase PageRank Model

The final aspect of our model is based on the observation that most traffic routes gradually move to higher throughput links from the origin, but the destination is usually not on these high throughput links. As it approaches a destination, the route tends to move towards lower throughput links. However, the memoryless aspect of PageRank models does not directly allow for a goal destination or these two phases.

Towards addressing this, we describe a two-phase PageRank model. Instead of a N state model, this becomes a $2N$ state model, where each link can be visited in the *up* phase as the route tends to move towards higher throughput links as it leave the origin, or in the *down* phase as the route tends to move towards lower throughput links as it approaches the destination. The up phase is now governed by a probability transition matrix $P^{\uparrow}$ with parameters $\alpha^{\uparrow} = (\alpha_s^{\uparrow}, \alpha_l^{\uparrow})$, and the down phase by a separate $P^{\downarrow}$ with its own parameters $\alpha^{\downarrow} = (\alpha_s^{\downarrow}, \alpha_l^{\downarrow})$. Then instead of a single parameter p to govern the probability the route ends, we need two parameters: β and ρ . The β parameter is the probability that when in the up phase the route transitions to the down phase, and ρ parameter is the probability that when in the down phase the route reaches its destination and ends. While after the down phase ends, the route needs to teleport to a new starting spot (based on $\tilde{E}_b$), when the up phase ends the route should stay on the same link as before. We can capture this entire in a transition probability matrix $M_b \in \mathbb{R}^{2N \times 2N}$, where each component is an $N \times N$ block:

$$M_b = \begin{pmatrix} (1 - \beta) P^{\uparrow} & \rho \tilde{E}_b \mathbf{1}^{\top} \\ \beta I & (1 - \rho) P^{\downarrow} \end{pmatrix}, \quad (7)$$

where I is the $N \times N$ identity and $\mathbf{1} \in \mathbb{R}^N$ is the all-ones column vector. With this convention the Markov chain's mass distribution $v \in \mathbb{R}^{2N}$ evolves as the simple matrix-vector product (which can be still implemented block-wise with efficient sparse operations):

$$v_{n+1} = M_b v_n. \quad (8)$$

Note that since state distribution v_n and the converged to $v_{\star}$ is now $2N$ -dimensional, in order to compare to a desired diffused population state $\tilde{E}_b \in \mathbb{R}^N$, we need to “fold” $v_{\star}$ into $\mathbb{R}^{2N}$. That is let $\hat{v}_{\star} \in \mathbb{R}^{2N}$ be defined for index $i \in [N]$ as $\hat{v}_{\star}[i] = v_{\star}[i] + v_{\star}[N + i]$. That is we add together the probabilities of being in the up state $i^{\uparrow} = i$ and the down state $i^{\downarrow} = N + i$.

Calibrating β and ρ . The two scalars β and ρ control how long the chain stays in each phase before switching. However, the sum of two geometric distributions follows a negative binomial distribution which does not match the observations in Figure 3; the negative binomial will have peak probability away from 0. We would rather like to interpret their ideal values as $\beta = \frac{1}{1 + \mathbb{E}[L_{\uparrow}]}$ and $\rho = \frac{1}{1 + \mathbb{E}[L_{\downarrow}]}$, where $L_{\uparrow}$ and $L_{\downarrow}$ are the empirical up- and down-phase path lengths. To do so, we need to model the lengths of the up and the down phase separately from the corpus.

Every matched route is a sequence of road links $(\ell_1, \ell_2, \dots, \ell_K)$ produced by the Viterbi pass of Section 3.1. For each trip we identify a single *peak link* ℓ_p as the link with the largest score $s(\ell) = \text{freespeed}(\ell) \times \text{numLanes}(\ell)$, with ties broken by the link closest to the trip's middle. The peak link plays the role of the trip's fastest, highest-capacity segment: the freeway core of a typical commute. The two phases of the chain are then defined as the prefix and the suffix of the route relative to this peak,

$$L_{\uparrow} = |\{j : j < p\}|, \quad L_{\downarrow} = |\{j : j > p\}|, \quad (9)$$

so $L_{\uparrow} + 1 + L_{\downarrow} = K$. By construction $L_{\uparrow}, L_{\downarrow} \in \{0, 1, 2, \dots\}$: a route can begin or end on its own peak link, in which case the corresponding phase has length zero. All 714,594 matched routes are therefore included in the calibration sample.

Figure 4 shows the histograms of $L_{\uparrow}$ and $L_{\downarrow}$ on this sample, with the fit geometric reference curve overlaid in black on each panel. Both empirical distributions are heavy at zero and decay roughly exponentially, which is the signature of a geometric distribution. The black curves overlay $\text{Geom}(p)$ with $p = \frac{1}{1 + \mathbb{E}[L]}$ using the empirical mean $\mathbb{E}[L]$. The empirical means are $\mathbb{E}[L_{\uparrow}] = 8.81$ and $\mathbb{E}[L_{\downarrow}] = 8.87$ for the up and down phases, respectively lead to $\beta = \frac{1}{1 + \mathbb{E}[L_{\uparrow}]} = 0.102$ and $\rho = \frac{1}{1 + \mathbb{E}[L_{\downarrow}]} = 0.101$. These estimated values $\beta, \rho \approx 0.10$, correspond to chains that stay in each phase about 10 steps on average.

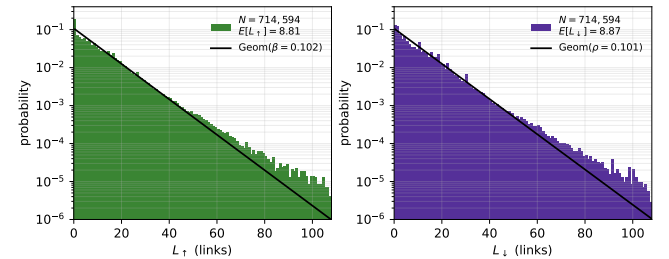


Figure 4: Empirical histograms of $L_{\uparrow}$ (left) and $L_{\downarrow}$ (right) on the full 714,594-trip matched corpus, with the matched geometric overlay in black.

Quantifying the fit. We pair the visual fit of Fig. 4 with the Kolmogorov-Smirnov statistic $D = \sup_k |F_{\text{emp}}(k) - F_{\text{geom}}(k; p)|$, where $F_{\text{emp}}(k)$ and $F_{\text{geom}}(k; p) = 1 - (1 - p)^{k+1}$ is the CDF of the fitted geometric. Smaller D means a tighter empirical match. Table 4 reports D in three configurations. Column D_{single} fits one geometric to the full route length $K = L_{\uparrow} + 1 + L_{\downarrow}$; columns $D_{\uparrow}$ and $D_{\downarrow}$ fit geometrics to $L_{\uparrow}$ and $L_{\downarrow}$ separately. The first row matches on the low-resolution map and aggregates to low-resolution links; the second matches on the high-resolution map and aggregates to the same links; the third both matches and aggregates at high-resolution. The two-phase model with the high-res matcher on the low-res links (that we use) has the smallest score, and is the best fit.

Table 4: Kolmogorov-Smirnov distance D between the single or two-phase histograms and the fitted geometric model.

Matcher	Granularity	D_{single}	$D_{\uparrow}$	$D_{\downarrow}$
low-res	low-res links	0.188	0.129	0.103
high-res	low-res links	0.129	0.030	0.032
high-res	high-res sub-edges	0.102	0.138	0.148

Fitting Two-Phase PageRank Model. Finally, to complete the two-phase PageRank model, we need to fit the parameters $\alpha^{\uparrow}$ and $\alpha^{\downarrow}$ for the full model to be complete. As in the single phase model, we set up a loss function

$$\mathcal{L}(\alpha) = \sum_{b \in [T]} \sum_{i \in [N]} \frac{|\hat{v}_{\star}(\tilde{E}_b, \alpha)[i] - \tilde{F}_b[i]|}{\tilde{F}_b[i] + \epsilon}$$

where $\hat{v}_{\star}(\tilde{E}_b, \alpha) \in \mathbb{R}^N$ now the folded, converged-to two-phase PageRank vector for a Markov chain following (8), where $P^{\uparrow}$ and $P^{\downarrow}$ are defined with the specified $\alpha^{\uparrow}$ and $\alpha^{\downarrow}$ parameters where now $\alpha = (\alpha^{\uparrow}, \alpha^{\downarrow}) = (\alpha_s^{\uparrow}, \alpha_l^{\uparrow}, \alpha_s^{\downarrow}, \alpha_l^{\downarrow}) \in \mathbb{R}^4$. Given the diffused starting positions $\tilde{E}_b$, the goal is still to match the diffused observed traffic population $\tilde{F}_b$. As before we choose these parameters $\alpha^* = \arg \min_{\alpha \in \mathbb{R}^4} \mathcal{L}(\alpha)$. We do this with gradient descent via autograd.

6 GLOBAL FORECASTING EVALUATION

We now evaluate our model on how well it predicts the global traffic distribution over the course of a week. We use MRE as described in Section 4.3. More specifically, we will aim to evaluate the model using the last 7 days of data. We report the Ov.MRE in three ways: the *train* setting evaluates on the first 23 days that are used to train α ; the *test* setting evaluates on the last 7 days, held out from the training of α , but this still uses the diffused starting point distribution $\tilde{E}_b$ to define teleportation for the same time period b ; and the *forecast* also evaluates on the last 7 days, but uses the diffused starting point distribution $\tilde{E}_{b-7d}$ from precisely 7 days prior; but against the diffused traffic population $\tilde{F}_b$ in the last 7 days. The first case train does not maintain a test/train split, and is shown for completeness. The test evaluation does maintain a test train split to evaluate the choices of α , but information may leak from the diffused starting points $\tilde{E}_b$ (which are different from the diffused traffic population $\tilde{F}_b$). Finally, the forecast version keeps all data from training part separate from the evaluation.

To be clear, in the forecast setting, we compute the average relative error for each link $i \in [N]$ for every b in the last week as

$$\text{MRE}_b = \frac{1}{N} \sum_{i=1}^N \frac{|\tilde{F}_{b,i} - v_{\star}(\tilde{E}_{b-7d}, \alpha)|}{\tilde{F}_{b,i} + \epsilon}.$$

The Overall MRE (Ov.MRE) is reported by considering 5 minute time bins $b \in [T-7d]$ for $T-7d = [2018-09-23 \text{ 19:00}, 2018-09-30 \text{ 18:55}]$ which are in the last week over observations, held out for evaluation. That is we compute and report

$$\text{Ov.MRE} = \frac{1}{T-7d} \sum_{b \in [T-7d]} \text{MRE}_b.$$

Table 5 shows the Overall MRE under various settings and baselines. The top half shows the Single Phase PageRank model, and the bottom half shows the Two Phase PageRank Model. The *uniform* row uses the uninformed teleportation process, where it starts a route at a uniform random link, each with probability $1/N$; it has no other tuning. The *origin dist* row replaces this with the diffused route origin prior from 7 days prior $\tilde{E}_{b-7d}$ (or $\tilde{E}_b$ for the train and test variations). This allows the model to adapt to the specific 5 minute time window b with the week. The remaining rows also use this prior $\tilde{E}_{b-7d}$ (or $\tilde{E}_b$), and also uses the training set T_{23d} to tune the various α parameters. While the last row *speed + lane* runs both α parameters, the other rows only tune a subset of them, with the others set to $\alpha_i = 0$ so that modeling has no effect on P .

Note that even the *uniform* (which uses no information from the data other than the graph structure) has higher Overall MRE on test than train. So while it is not surprising for test scores to be higher than training scores (due to perhaps slight overfitting on the training set) this difference reflects that the inherent harder nature of the last 7 days, in the evaluate split we chose.

Otherwise, we observe that from the uniform baseline, as we add enhancements (origin distribution $\tilde{E}_b$, training α , two-phase), the Ov.MRE error decreases. For each measurement variant, the (highlighted) row which includes all of these enhancements has the lowest error. As predicted the $\alpha^{\uparrow}$ values optimize to be positive for the up phase, and the $\alpha^{\downarrow}$ values optimize to negative for the down phase.

We remark that the decrease in the error may not seem significant; with relative error about 1 after optimization. We posit that this sort of error range is expected and it would be surprising to improve upon this. If we similarly measure Ov.MRE of $\sum_{i \in [N]} \frac{|\tilde{F}_b[i] - \tilde{F}_{b-7d}[i]|}{(\tilde{F}_b[i] + \epsilon)}$ (that is, the diffused population from one week to the next) averaged over a full week, we obtain about 0.67, which is on the same order of our error. Moreover, if we use the non-diffused data F_b vs. F_{b-7d} , then the Ov.MRE is around 9.7, much less than our error. Hence there is significant variation between weeks, and we achieve less than twice this on diffused data, and far less on raw data. Also note that the standard deviation is relatively large (about 0.8) and also decreases as we progress to more enhanced models. Overall, this shows that our methods provide real improvement in its week-to-week modeling ability, which generating memoryless routes on the full metro-scale network.

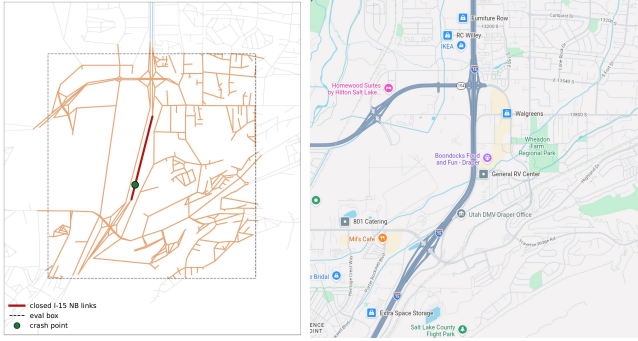
7 EVENT MODELING

We show how to apply this framework to events that have predictable configuration changes, which allows for updated modeling.

There are two direct ways we can leverage our model. First, if there are road closures (either planned or unplanned), then we can perform a simple modification of the graph by removing those links. Second, if there is a large event attracting many of additional people to a small area, then we can adapt the route origin distribution $\tilde{E}_b$ for the specified time windows b for the event. In practice for sporting events or festivals we can use prior events for these modified distribution estimates, but because we only have access to one month of data, we use a training subset of the data from the

Table 5: Overall MRE and standard deviation for PageRank models, evaluated on train data (first 23 days T_{23d}), test data (last 7 days T_{-7d}), and forecast where starting point distribution $\tilde{E}_{b-7d}$ is from 7 days prior.

Phases	mode	α_s or $\alpha_s^\uparrow$	$\alpha_s^\downarrow$	α_l or $\alpha_l^\uparrow$	$\alpha_l^\downarrow$	train	(st.dev)	test	(st.dev)	forecast	(st.dev)
						$(\tilde{E}_b, \tilde{F}_b); T_{23d}$		$(\tilde{E}_b, \tilde{F}_b); T_{-7d}$		$(\tilde{E}_{b-7d}, \tilde{F}_b); T_{-7d}$	
1	uniform: $E = 1/N$	—	—	—	—	1.1739	(0.8309)	1.2706	(0.8625)	1.2706	(0.8625)
1	origin dist	—	—	—	—	1.1266	(0.7818)	1.2178	(0.8117)	1.2401	(0.8310)
1	speed	+2.242	—	—	—	1.1096	(0.7466)	1.1963	(0.7759)	1.2197	(0.7960)
1	lane	—	—	+0.673	—	1.1136	(0.7586)	1.2018	(0.7884)	1.2252	(0.8086)
1	speed + lane	+1.751	—	+0.436	—	1.1045	(0.7383)	1.1903	(0.7675)	1.2141	(0.7879)
2	uniform: $E = 1/N$	—	—	—	—	1.1732	(0.8309)	1.2698	(0.8625)	1.2698	(0.8625)
2	origin dist	—	—	—	—	1.1211	(0.7768)	1.2117	(0.8065)	1.2365	(0.8279)
2	speed	+6.960	-1.518	—	—	1.0990	(0.7446)	1.1852	(0.7739)	1.2117	(0.7966)
2	lane	—	—	+2.320	-0.658	1.1034	(0.7490)	1.1902	(0.7786)	1.2176	(0.8020)
2	speed + lane	+4.649	-0.572	+1.161	-0.167	1.0948	(0.7375)	1.1801	(0.7667)	1.2072	(0.7899)

**Figure 5: I-15 northbound crash near Draper, 2018-09-20: the closed link in red, the evaluation box dashed, and the same area on the basemap.**

$\tilde{E}_b$ data actual event. In both cases, we do not need to retrain the α parameters, only to rerun the PageRank iteration (8); which is fast because we maintain a sparse format, and the distribution vector $v_\star$ is converged to quickly.

Moreover, on a metro scale network, even large events have relatively local effect. As a result, we change the model prediction and evaluate their effect on local region around the events (in addition to the global evaluation – which barely changes).

7.1 Case study 1: Car Crash

On Thursday, September 20, 2018 a car crashed on the I-15 northbound, which closed a freeway segment during the morning. We run our model over the 60 five-minute slots from 07:00 to 11:55 and report the Overall MRE on an evaluation box around the closure (975 open links) and on the whole network. The teleportation prior is the previous week’s diffused origins $\tilde{E}_{b-7d}$ (the same hours on Thursday, September 13) and the target is the diffused crash-day population $\tilde{F}_b$; we keep our model parameters fixed. The closed region is the single northbound I-15 link covering the crash point.

We compare the unmodified network (*no surgery*) with *surgery*, which deletes the closed link from the chain’s transition matrices,

Table 6: I-15 Draper crash: Overall MRE with and without surgery, under the previous-week prior.

condition	eval box	whole network
	OV. MRE (st. dev)	OV. MRE (st. dev)
no surgery	1.8475 (0.7605)	1.7374 (0.7259)
surgery	1.8214 (0.7558)	1.7395 (0.7269)

so that no walk can step into or out of it, and re-solves for $v_\star$ on the reduced network. Table 6 (previous-week prior) reports the results. With only one freeway link closed, surgery shifts the box error modestly (about 1–2%).

7.2 Case study 2: Large Sporting Event

A Saturday football game at Rice-Eccles Stadium (Washington at Utah, 2018-09-15, kickoff 20:00, attendance 47,445) is a pure demand event: it closes no road, but it concentrates trips around the venue. We therefore keep the full network and intervene only on the prior. We score on an evaluation box centered on the stadium (Fig. 6; 1,367 links) over the 12 five-minute bins from 23:00 to 23:55, the end-of-game departure window (the game ended at 23:19), comparing the chain output to the game-day diffused traffic population. The fed prior is the previous week’s diffused origins (the clean Saturday 2018-09-08, same hours).

Normally, we would learn the expected traffic from similar prior events, but without that in the dataset, we perform a test train split on the links within the evaluation box, and learn the increase in traffic population that way. To do so, we partition all links into the evaluation box B and its complement A , with prior masses $a = \sum_{x \in A} p(x)$ and $b = \sum_{x \in B} p(x)$, $a + b = 1$. We move a traffic probability ψ from A into B :

$$q(x) = p(x) \left(1 - \frac{\psi}{a}\right) \quad (x \in A), \quad q(x) = p(x) \left(1 + \frac{\psi}{b}\right) \quad (x \in B).$$

This keeps q a probability distribution ($\sum_x q(x) = 1$) and stays non-negative for $-b < \psi < a$. A positive ψ concentrates the fed demand

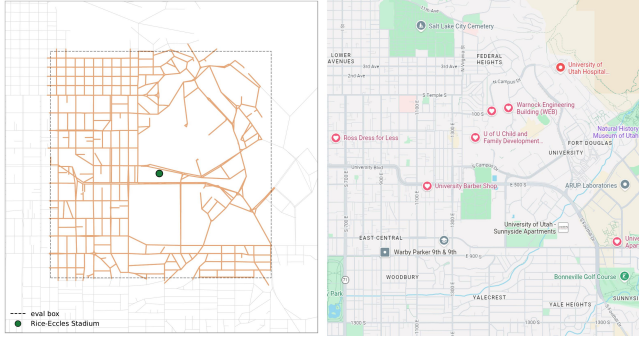


Figure 6: Rice-Eccles study area: the evaluation box (dashed) and the stadium (marker); no links are closed.

Table 7: Stadium: learned population-transfer ψ^* with train and held-out test eval-box mean squared error (single random 75/25 split of the box links). The last column is the held-out test error at $\psi = 0$.

model (ψ^*)	train	test	$\psi=0$
SP (+0.012)	6205.55	6326.10	6405.14
TP (+0.011)	6222.20	6327.89	6403.54

on the stadium box, a negative ψ thins it, and $\psi = 0$ recovers the unmodified previous-week prior; the shape within each block is unchanged, only the box's share of the mass moves.

We treat ψ as the single free parameter and choose it to minimize the eval-box mean squared error on a training portion of the data, then report the error on a held-out portion. We validate it over the 1,367 box links (random 75% train, 25% test), with a fixed seed; the single-phase and two-phase models are fit independently.

Table 7 gives the learned ψ^* and the held-out box mean squared error. Both models learn a positive transfer ($\psi^* \approx +0.012$) that concentrates the fed demand on the stadium box, and it lowers the held-out box squared error below the unmodified previous-week prior. A single, data-light scalar fit on a subset of links thus transfers cleanly to the held-out remainder.

7.3 Case study 3: Street festival

The 9th & 9th Street Festival (Saturday 2018-09-15, 10:00 to 17:55) closes 900 South and 900 East at the 9th-and-9th intersection, 14 links in all (Fig. 7). Unlike the stadium, the festival removes links, so we first apply surgery (as with the crash, §7.1): we remove the 14 closed links from the network, zero them out of the prior, and redistribute their popularity onto the evaluation box, re-solving the chain on the open network. On top of surgery we intervene on the prior. As in the stadium study (§7.2), we feed the previous week's diffused origins (the clean Saturday 2018-09-08, same hours) and score on the evaluation box (Fig. 7; 694 open links) over the 12 five-minute bins from 17:00 to 17:55, the closing hour of the festival, comparing the chain output to the festival-day diffused traffic population.

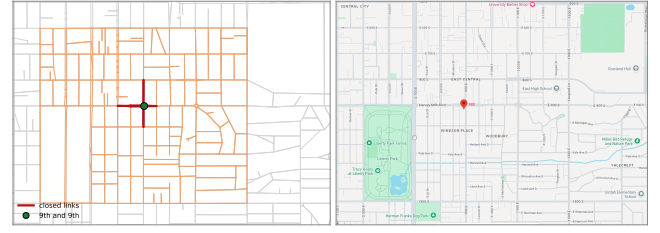


Figure 7: Street festival study area: the 14 closed links (red), the evaluation box (dashed), and the same area on the basemap.

Table 8: Festival: learned prior transfer ψ^* on top of surgery, with train and held-out test eval-box mean squared error (single random 75/25 split of the 694 box links). The last column is the held-out test error at $\psi = 0$.

model (ψ^*)	train	test	$\psi=0$
SP (+0.005)	4306.58	5285.57	5410.22
TP (+0.005)	4314.97	5321.71	5432.69

We apply the prior mass-transfer ψ of §7.2 on top of surgery, moving a traffic probability ψ from the open complement into the evaluation box. We treat ψ as the single free parameter and choose it to minimize the eval-box mean squared error on a training portion of the box links (random 75% train, 25% test, fixed seed), then report the error on the held-out portion; the single-phase and two-phase models are fit independently.

Table 8 gives the learned ψ^* and the held-out box mean squared error. Both models learn a positive transfer ($\psi^* \approx +0.005$) that concentrates the fed demand on the festival box, and on top of surgery it lowers the held-out box squared error below the unmodified prior. A single, data-light scalar fit on a subset of links thus transfers cleanly to the held-out remainder.

HAMID: unfortunately two-phase doesn't work better than single phase here.

8 CONCLUSION AND DISCUSSION

We show how to build a metro-scale traffic model at the link level, at 5-minute resolution, from large yet extremely sparse data. Such data is very noisy week to week, and overcoming the noise and sparsity is a significant challenge.

A key feature of our model is that it is a learned PageRank model, that is optimized to the training data. This means that the model can not only easily generate full routes, but also quickly provides full metro-wide traffic population estimates. Competing methods do not offer this level of resolution for the routes, or do not allow for directly inferred population predictions.

Through a combination of realistic traffic smoothing and careful modeling, we produce estimates that are plausible and significantly better than the raw measurement distributions. We believe this underexplored modeling at this scale offers a lot of promise for metro-scale foundation models for traffic and mobility.

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APPENDIX

Table 9 shows similar statistics for the cells with initial counts of 0. In this case it only makes sense to use the global noise floor τ_{global} . The median and 95th-percentile ratios are essentially constant at $\approx 4 \times 10^{-4}$ across all γ : at least 95% of zero cells never see diffusion mass, they only carry the Laplace floor α/N . The fraction of zero cells exceeding the half-noise tolerance is 0.08% at $\gamma = 0.20$ ($\sim 1.5 \times 10^5$ cells out of 1.95×10^8); these are the cells immediately adjacent to active links and they are the ones that get filled in. The takeaway is that the diffusion is bounded even on the cells where the per-cell criterion of Table 3 is undefined: it lifts a small adjacency-defined tail of zero cells into the noise band and leaves the rest at the Laplace floor.

Table 9: Displacement of zero cells ($n_{\text{zero}} = 1.95 \times 10^8$) in units of $\tau_{\text{global}} = 2.65 \times 10^{-4}$.

γ	median	95th pct.	> 0.5
0.00	0.0004	0.0004	0.00%
0.05	0.0004	0.0004	0.01%
0.10	0.0004	0.0004	0.03%
0.12	0.0004	0.0004	0.04%
0.13	0.0004	0.0004	0.04%
0.14	0.0004	0.0004	0.05%
0.15	0.0004	0.0004	0.05%
0.20	0.0004	0.0004	0.08%
0.25	0.0004	0.0004	0.10%
0.30	0.0004	0.0004	0.13%
0.40	0.0004	0.0004	0.20%
0.50	0.0004	0.0004	0.28%
0.70	0.0003	0.0004	0.46%

Table 10

Hour	Model	ψ^*	Train@ ψ^*	Train@0	Test@ ψ^*	Test@0
18	SP	+0.0060	4098.92	4116.94	3199.26	3215.43
18	TP	+0.0040	4119.03	4130.26	3208.71	3220.75
19	SP	+0.0180	7789.84	7967.86	6729.54	6974.11
19	TP	+0.0160	7832.43	7985.14	6783.59	6989.91
20	SP	+0.0080	5608.02	5648.78	5262.27	5312.08
20	TP	+0.0070	5634.68	5663.83	5296.18	5330.86
21	SP	+0.0020	2967.42	2970.00	3609.55	3614.71
21	TP	+0.0010	2978.30	2979.32	3623.94	3626.01
22	SP	+0.0020	2974.83	2976.73	4548.24	4560.84
22	TP	+0.0010	2983.92	2984.89	4565.62	4571.71
23	SP	+0.0120	6205.55	6281.02	6326.10	6405.14
23	TP	+0.0110	6222.20	6289.60	6327.89	6403.54